

Self-sensing SMA bending wire

1. Introduction

In the first three documentation I neglected a fact which is temperature sensing is not an easy task, as I didn't consider that casual thermocouple has response time 230 ms (which is not by any means good for such sensitive actuator as any error in temperature would have enormous effects on the output whether it's curvature, strain or angular position, etc. In this documentation I will develop self-sensing method utilizing the fact the wire resistance change in the different phases, that little fact is the major drive to this project.

2. The physical parts of the system

1. 2 thermocouple(MAX6675) it's response time 230 ms
2. A current sensor it's response time 1 ms
3. A 3 V DC power supply (safety measure)
4. N-type MOSFET

Utilizing those parts I will build self-sensing method that will act as the feedback sensor in the inner loop temperature controller, for the thermocouples they will be used in two vital parts in the experiment, one measuring the room temperature frequently, the other one will be used in the calibration of the temperature self-sensing method by taking the reads every 230 ms, the current sensor will help in calculating the varying resistance consequently I will be able to measure the electric power, for the power supply a 3V DC voltage will be safe enough and we could go lower than that in calibration in specific, N-type MOSFET to avoid excessive currents.

3. The multi-rate calibration and validation

As I said in the introduction the variable resistance of the shape memory alloy would help us to estimate the temperature, this variable resistance is the reflection of the shape memory effect knowing this fact we can calculate the martensite volume fraction by easy computation, the SMA have two extreme resistances R_m which is martensite phase resistance, the R_a which is austenite phase resistance, getting these resistances is the first step in our experiment, with this equation we can drive the martensite volume fraction as the equation(1) implying I will only be able to calculate the martensite volume fraction if we get the two resistances of the phases when the wire is totally martensite or totally austenite.

$$R_{measured} = R_a + \xi(R_m - R_a) \quad (1),$$

Step (a) calculating the martensite resistance (R_m) ($R_{measured} = R_a + 1(R_m - R_a)$)

By applying very small voltage to excite a very small current that can't heat the wire to austenite phase we'll be able to get R_m by Ohm's law $measured = 1(R_m - 0)$.

$$R = V/I \quad (2)$$

Step (b) calculating the austenite resistance (R_a) ($R_{measured} = R_a + 0(R_m - R_a)$)

We can apply a load or deform the wire and by heating it will be back to its original shape and we can calculate the austenite resistance at that time, I preferred isolating the two step rather than making it in one step for accuracy.

Step (c) computing the martensite volume fraction

By obtaining these resistances and as I calculate the real-time resistance I can compute the martensite volume fraction in real time also with its change of rate which is used in computation the latent heat part.

Step (d) manufacturing of the observer

The observer takes the measured resistance if there is a change it will estimate the temperature with the latent heat included in its equation, if not it will estimate the temperature without it included in the equation as the rate of change of the martensite volume fraction is zero, It will also compute the changing electrical power, then there come the three adjustable parameters h , C_p , ΔH , these three parameters can only be guessed where they can really be as for coefficient of convection in still air it will be in range 15 to 30 $W/m^2.K$, for thermal capacity from 320 to 450 $J/Kg.K$ and for latent heat constant ΔH it can be in range 16000 to 32000 J/Kg .

$$m C_p (dT/dt) = I^2 R(\xi) - h A_s (T - T_\infty) - m \Delta H (d\xi/dt) \quad (3)$$

Step (e) the multi rate calibration

The thermocouple I mentioned before has 230 ms and the current sensor has 1 ms if my experiment runs for 3 seconds I have 3000 samples of the current sensor and only 14 samples from the thermocouple which is measuring the actual temperature of the wire (in slow rate) by doing this we can now adjust h and C_p to reach a better accuracy.

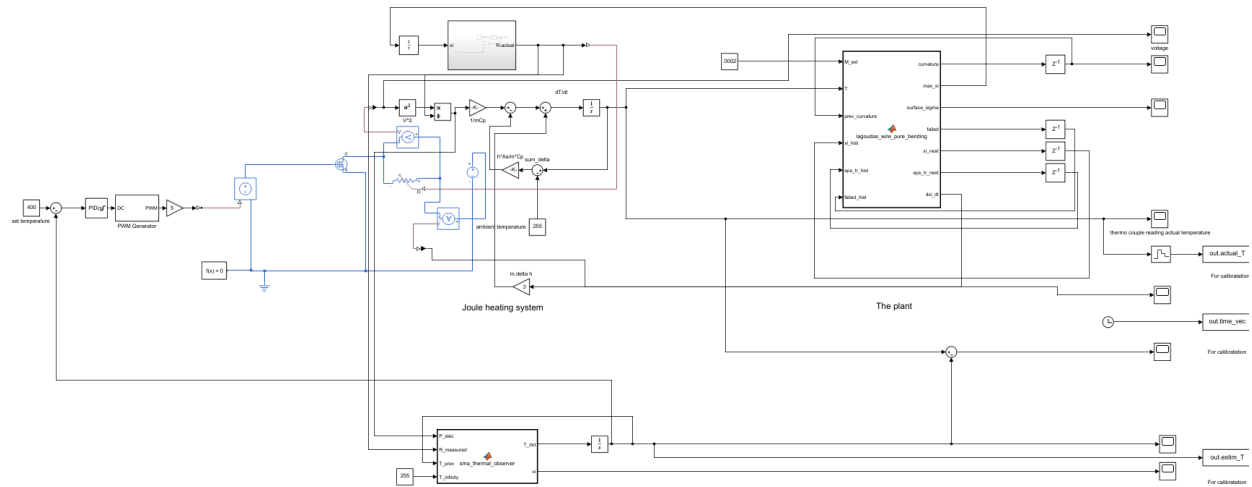


Figure (1) the system setup

We set temperature to reach 300 K using the observer estimated temperature as feedback, the observer has different parameter values of h , C_p and ΔH to Joule heating system, we will run the simulation at first with zero load for calibration then validate the results at zero load and load applied situation.

Part (1) calibrating and validation at zero load

Step (a) extracting the martensite phase resistance

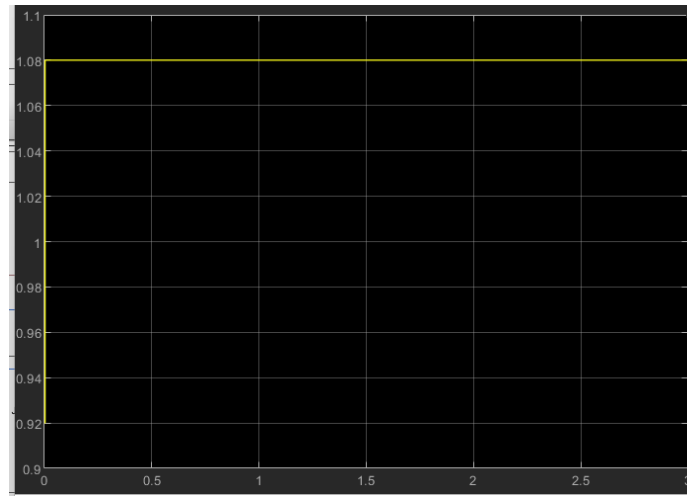


Figure (2) the measured resistance plot

The plot shows that the resistance is at 1.08 ohm which is the martensite phase resistance.

Step (b) extracting the austenite phase resistance

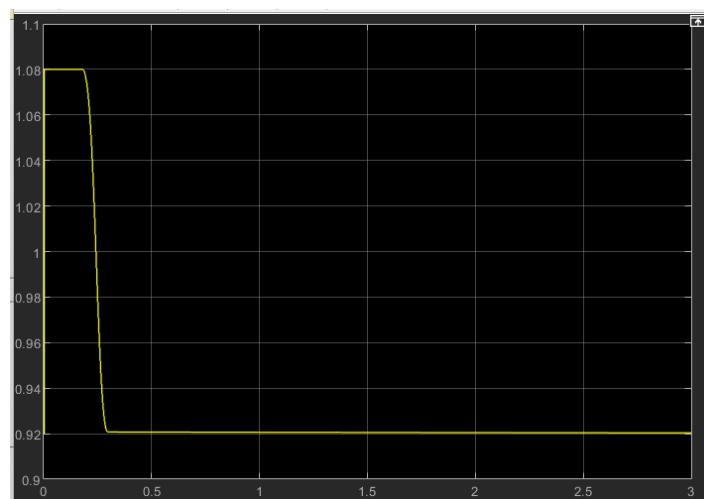


Figure (3) the measured resistance curve in heating

We see the measured resistance decreased as the wire went from martensite phase to austenite phase to .92 ohm which is the austenite phase resistance.

Step (c) computing the martensite volume fraction

As we have got from step (a) and step (b) the required resistances we can now compute the martensite volume fraction as implied in equation (1).

Step (d) constructing the observer

Since the observer parameters can't match the exact real electro-thermal part of the wire I will assign them different from the driving electro thermal system by changing the coefficient of convection h , the heat capacity Cp , the latent heat coefficient ΔH they are assigned, $15W/m^2.K$ (as the real application will act in still air), $400 J/Kg.K$, $24000 J/Kg$ the values is what represents the real electro-thermal system.

Step (e) multi-rate calibrating

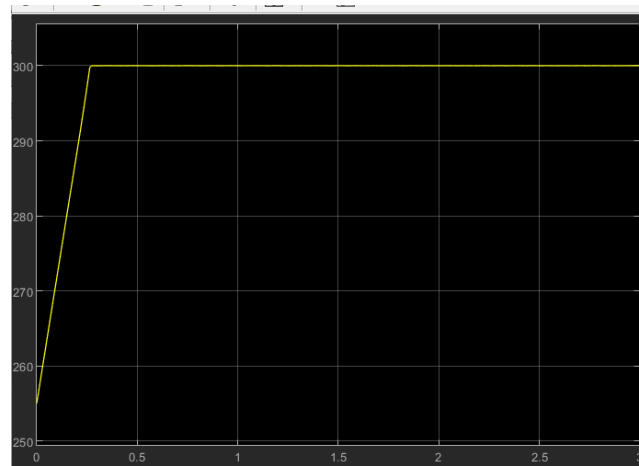


Figure (4) the estimated temperature curve

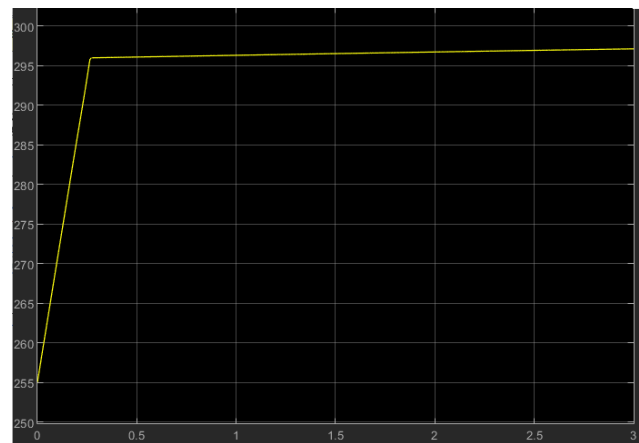


Figure (5) the actual temperature curve

In the real application the estimated curve figure (4) is based on $1 ms$ response time of the current sensor as we can't get the actual temperature curve in real life in example if I have 3 second experiment we have 3000 readings for estimated temperature to 14 readings for the thermocouple.

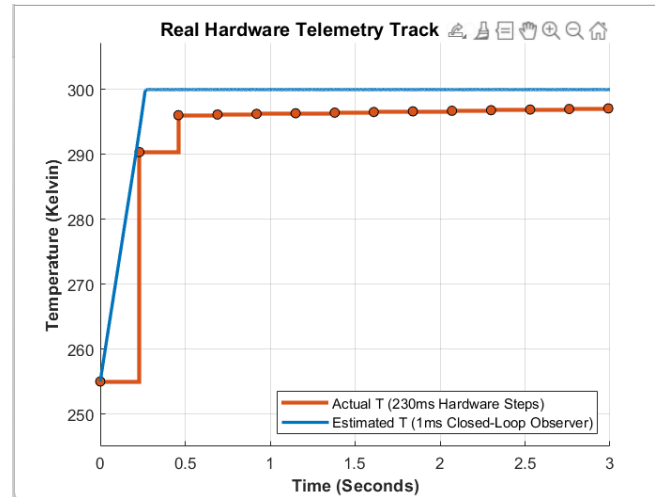


Figure (6) the 14 samples of the thermocouple vs estimated temperature curve

So I mimicked the real life thermocouple readings and by comparing, and analyzing the 14 sample curve in comparison to the estimated curve, for sample one and two of the thermocouple are located on the estimated curve from sample 3 to 14 we can notice is the curve is under damped and have increasing drift as time goes which is can't be tolerated on SMA actuation, now we have two parameter to adjust h and C_p cause they are existing in every phase of the wire (transition phase or fully shifted phase) I prefer here to manually adjusting them rather than resorting to parameter estimation method since we can easily deduce laws that can help us to get a better result for the case I will increase the C_p parameter dealing with underdamped issue of the curve and decrease the h parameter to treat the increasing drift.

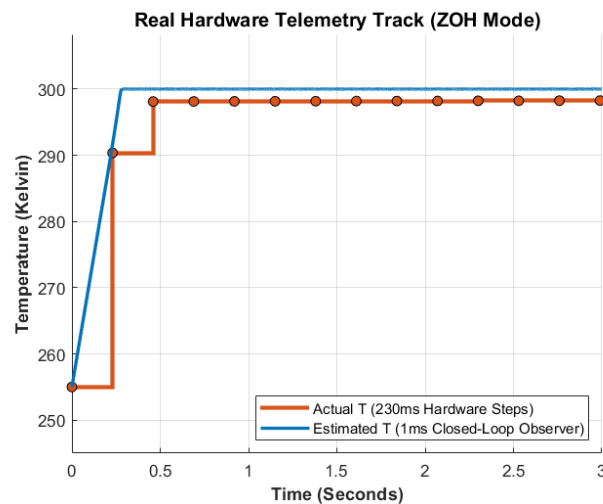


Figure (7) the estimated temperature and the thermocouple samples curve after the first update of the parameter

We see here we cured the increasing drift by decreasing h but there is still underdamping problem so I increase C_p to match the steady state curve.

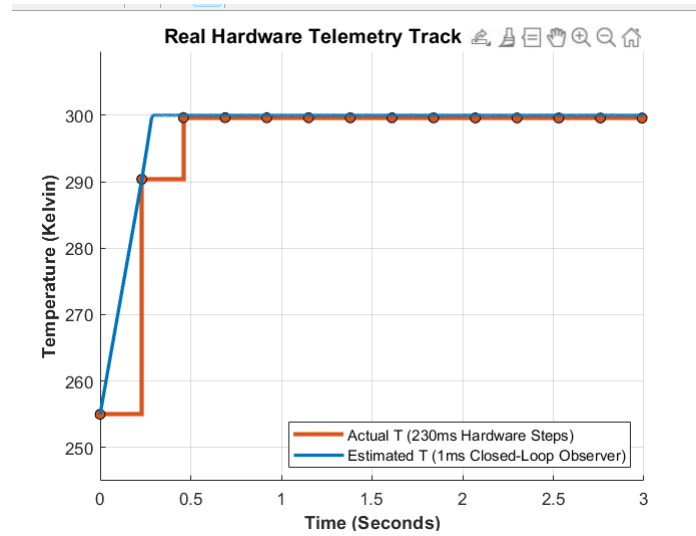


Figure (8) the estimated vs actual temperature curves after increasing C_p

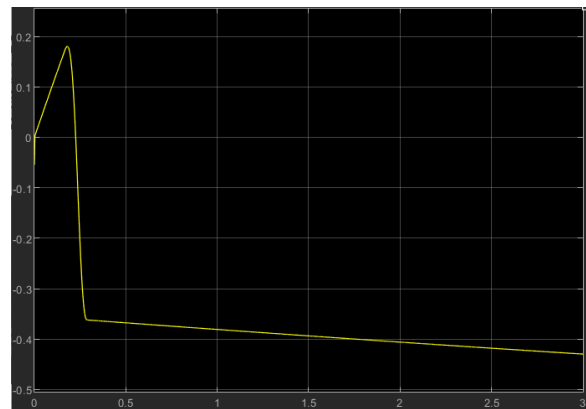


Figure (9) the error between the actual temperature and estimated temperature

Here we notice there is a small drift after .35 seconds but it's a tiny value and the fact that it is decreasing can't change the position of the wire since the temperature should decrease to martensite phase to change the wire position but it will change the position of the wire if it is increasing but I want to get a better and stable result by increasing h by a small value and test the laws I deduce.

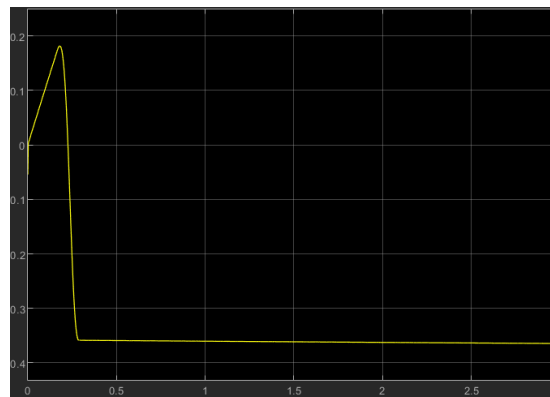


Figure (10) the error between the actual temperature and estimated temperature after a small increase in h parameter

Here the error between the estimated temperature and the actual one is turned to be stable and steep drift got dealt with

Part (2) the validation of the self-sensing at different cases

In this part I will test the new calibrated observer parameter in different load and different input voltages

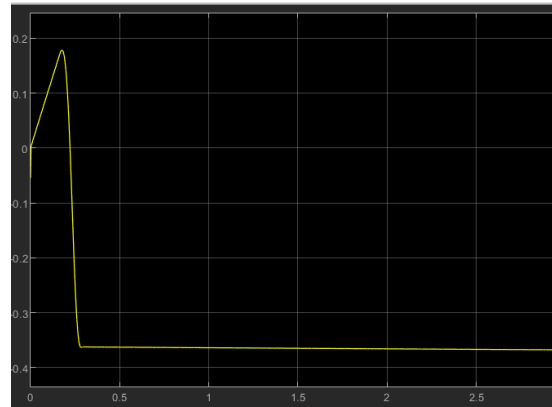


Figure (11) the error between the estimated temperature and the actual temperature in load applied case

Figure (11) shows no change in load applying case.

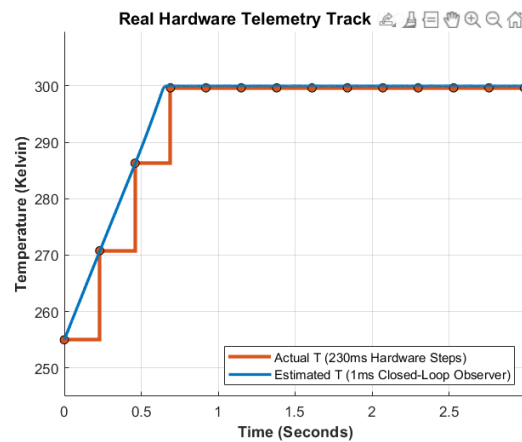


Figure (12) the estimated temperature vs the sampled actual temperatures curve with a 2V input Voltage

Figure (12) shows no difference with different input voltage than the original 3V input voltage.

4. The overall system robustness

This section is dedicated to test the overall system reaction with error in temperature estimation, insufficient FORC measurement samples with not 100 % accurate temperature/curvature data and with loads that are different than The FORC baseline load, first I made the observer to have an error in temperature estimation deliberately, afterward I'm going to collect FORC data with this estimation error at some load, then test the system at different load than this load

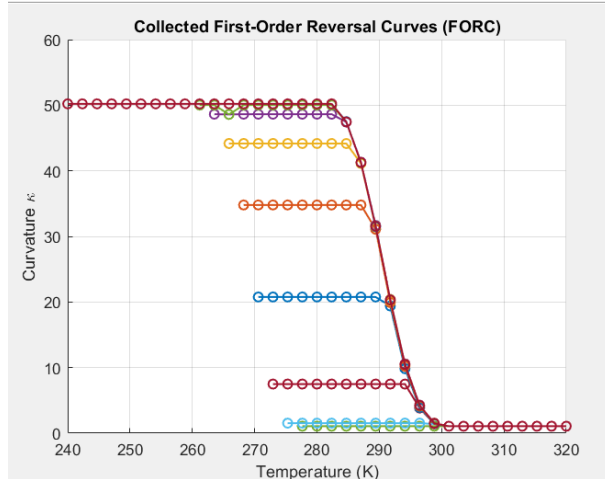


Figure (13) The 35 sample FORC

Notice the gaps in the Figure (13) which means the hysteresis model of the system is not totally captured.

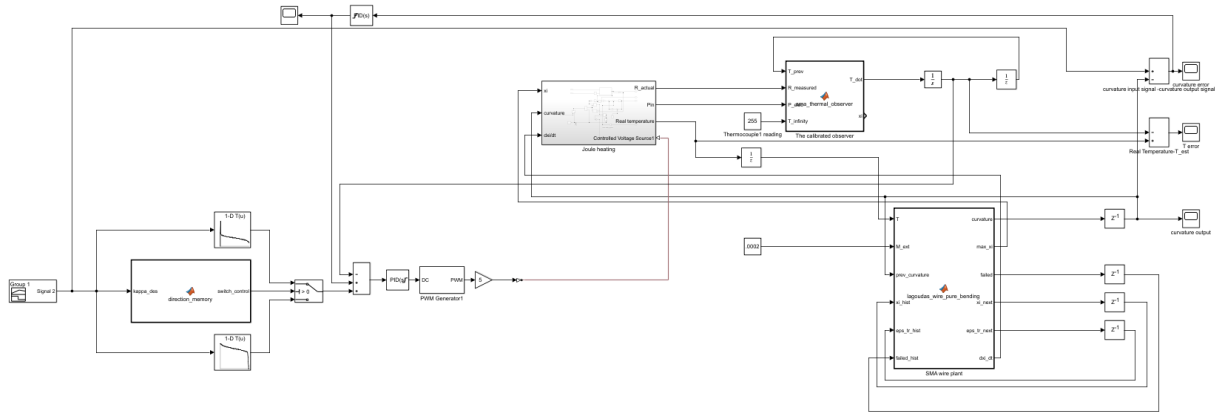


Figure (14) The hybrid feedforward controller with temperature observer

Case (a) The system reaction at baseline load

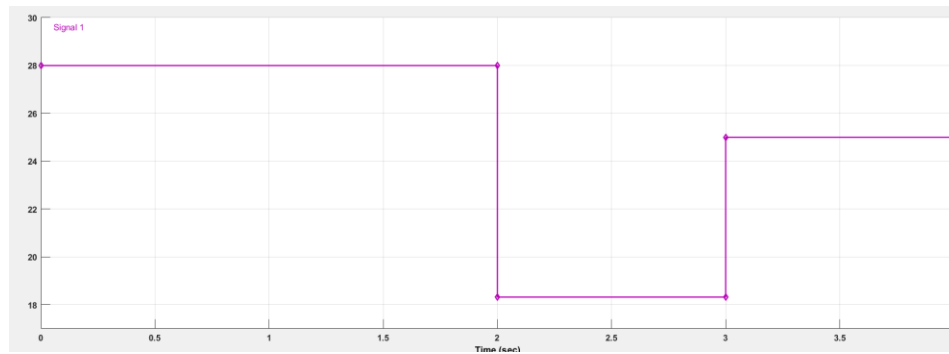


Figure (15) The input signal

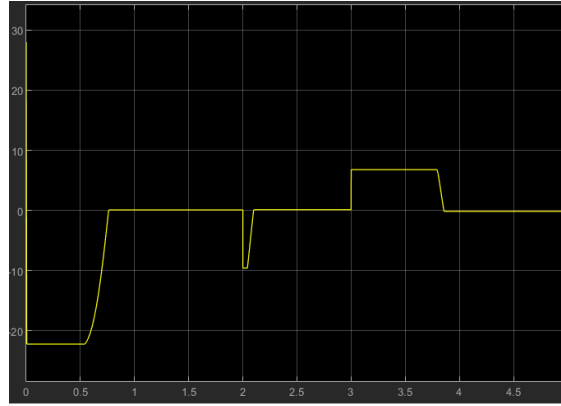


Figure (16) The curvature error plot

The plot shows the steady state error is zero and the plateaus in the system is governed by the applied voltage and the convection method which for the sake of increasing the testing speed and FORC extraction (It took hours with even the gain of the liquid cooling) I increased it to be $515 \text{ W/m}^2 \cdot K$ and 2V as applied voltage.

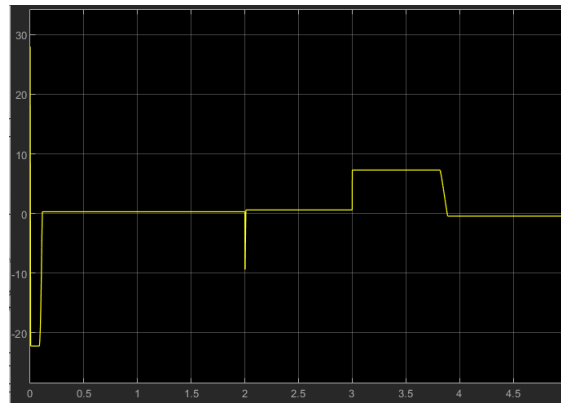


Figure (17) The curvature error plot with 4.5V as input voltage

With the same auxiliary PI controller gains, I increased the applied Voltage 4.5 with the same inner loop temperature controller PI parameters we removed the plateaus of the heating path.

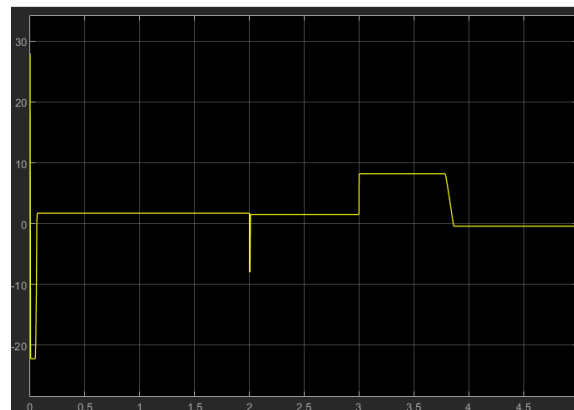


Figure (18) The curvature error plot with 6V as input voltage

Figure (18) show a steady state error at heating steps, the reason could be the 2V self-sensing calibration or the FORC didn't capture this part, or the combination of both problems, but we can work with 4.5V if we need to speed up the heating steps it's not that bad just a milliseconds difference with a better control output

Case (b) The system reaction to a load higher than the baseline

This part I will test different signals with different applied voltage in a higher load case than the baseline

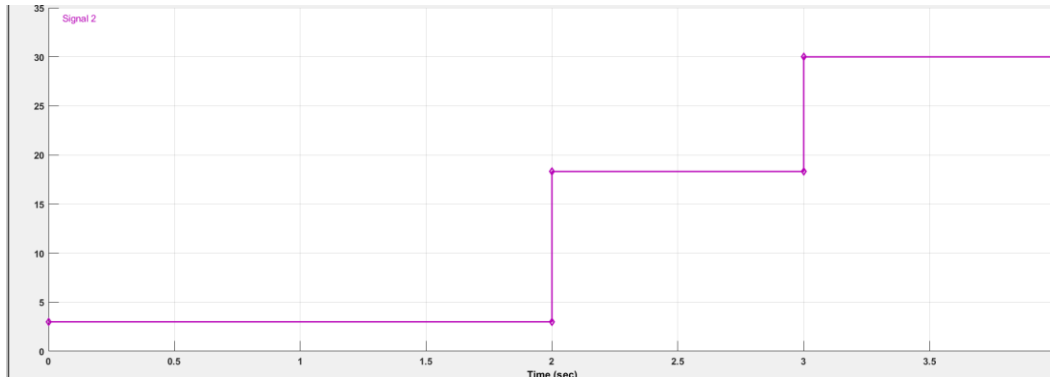


Figure (19) the new testing trajectory

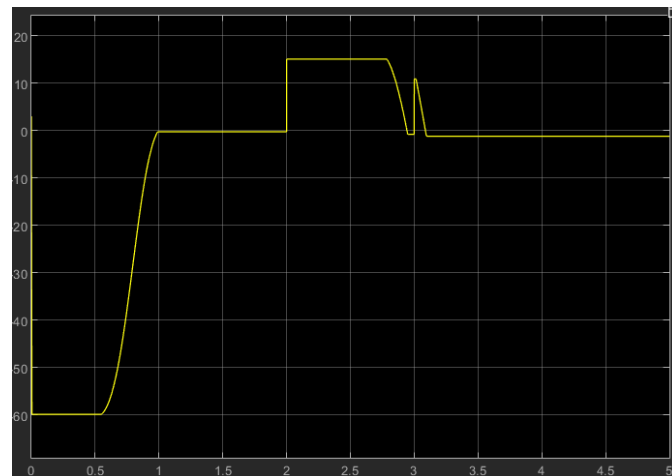


Figure (20) The curvature error plot with 2V applied voltage

After changing the input trajectory with 2V applied voltage and a higher load than the baseline, we have a small steady state error at the cooling step.

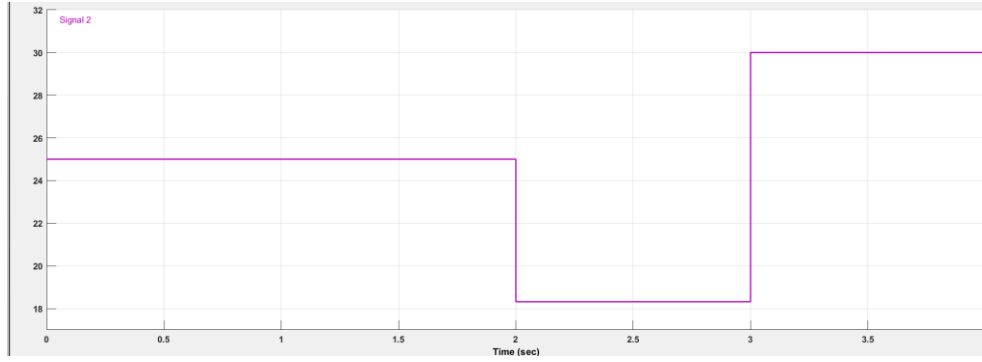


Figure (21) A new desired trajectory

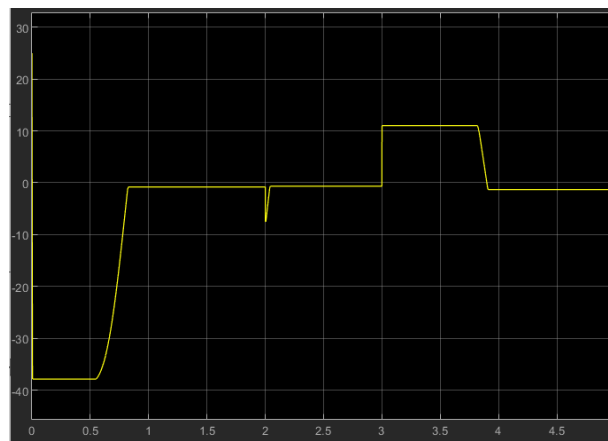


Figure (22) the curvature error plot of new desired trajectory with 2V voltage applied

A steady state error in both cooling and heating steps but it $-1\ m^{-1}$ heating step and $-3m^{-1}$ cooling step.

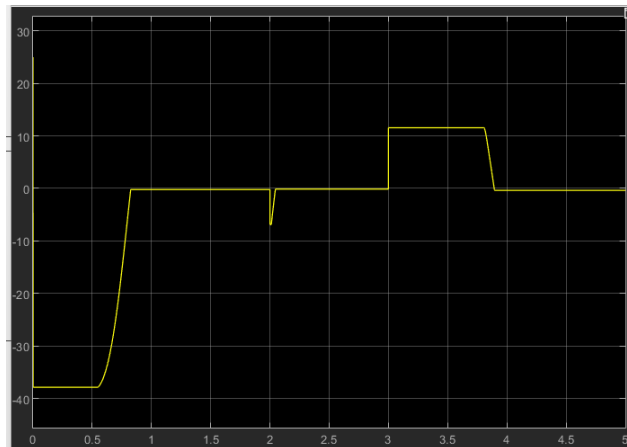


Figure (23) The curvature error plot after the auxiliary PI controller gain tuning

Figure (23) the steady state error is decreased at both cooling and heating steps.

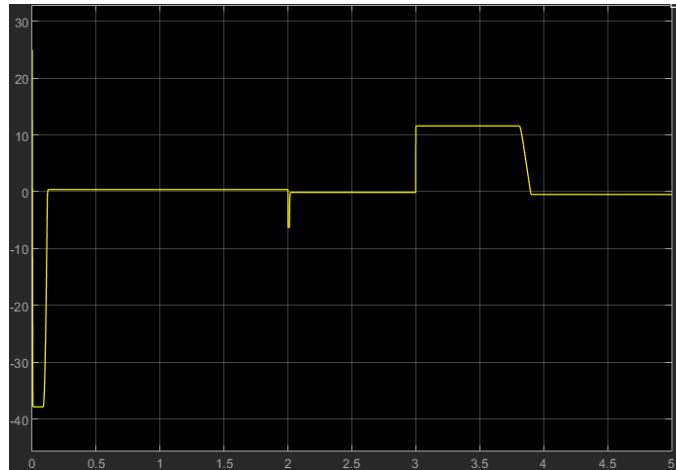


Figure (24) the curvature error plot with 4.5V applied voltage

Figure (24) shows a slight steady state error at first heating step.

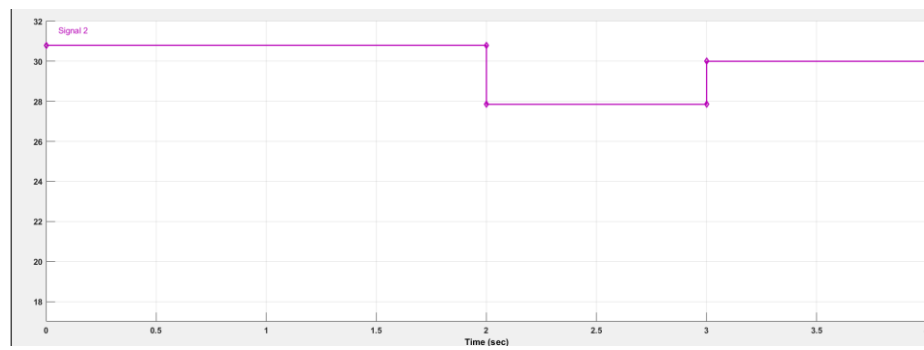


Figure (25) New testing trajectory

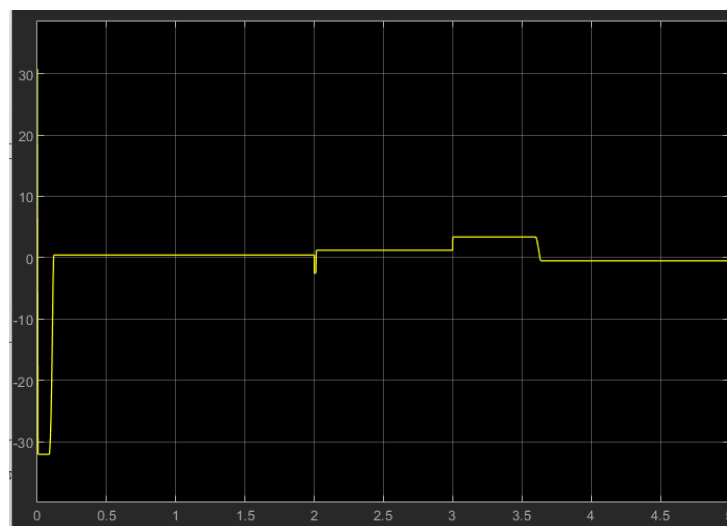


Figure (26) The curvature error plot

Figure (26) shows a slight steady state error but we can't guess where it come from exactly whether it is the lack of FORC samples or the temperature estimation error but as long as it is in the range is tolerated or I will be trapped in endless loop of PI gain tuning.

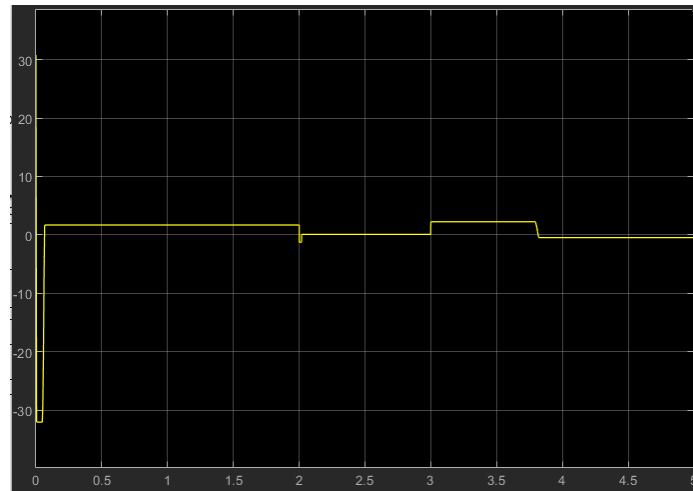


Figure (27) The curvature error plot with 6V applied voltage

The same error of the baseline as the baseline experiment in the first step it seems the cause is that we drifted away the 2V calibrating this occurs so the conclusion of this varying applied voltage is we should stick around 2V or the calibration applied voltage, also it seems 4.5V applied voltage is the best choice in terms of stability and heating speed.

Case (c) the system reaction with lower load than the baseline

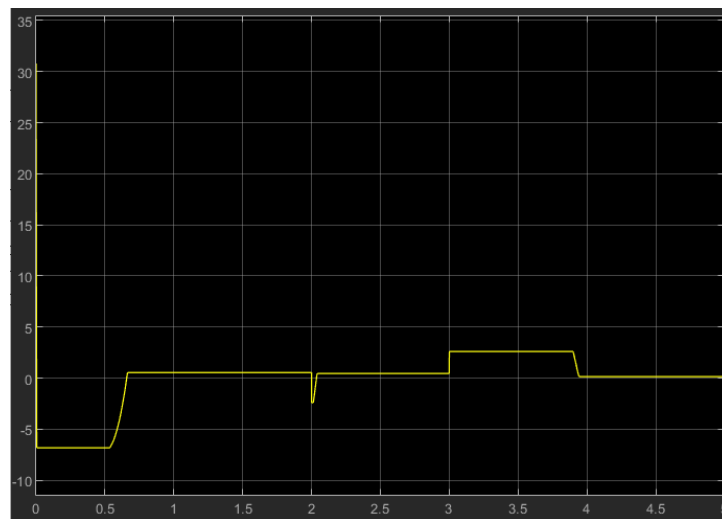


Figure (28) The curvature error plot with 2V applied voltage

The steady state error is near zero in both heating and cooling steps with 2V applied voltage.

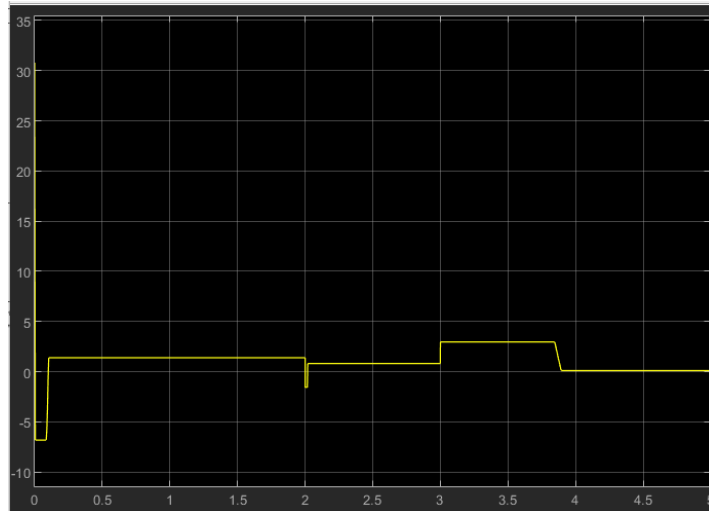


Figure (29) The curvature error plot with 4.5V applied voltage

Figure (29) shows a slight increase in the steady state error in the first step about 1 m^{-1} and no change at cooling step so it's in tolerated range.

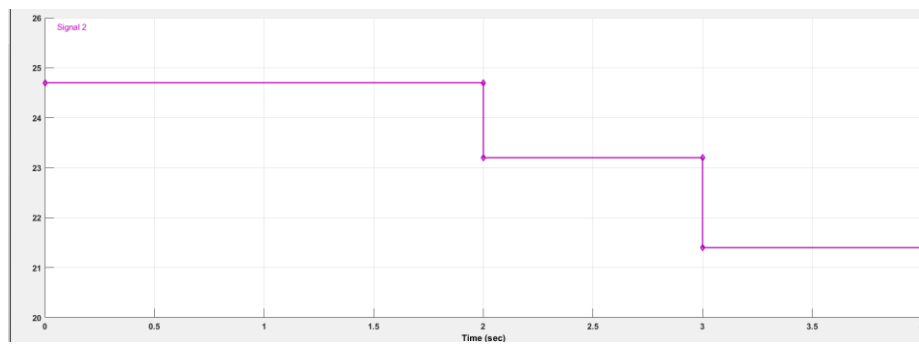


Figure (30) New testing trajectory at 4.5V applied voltage

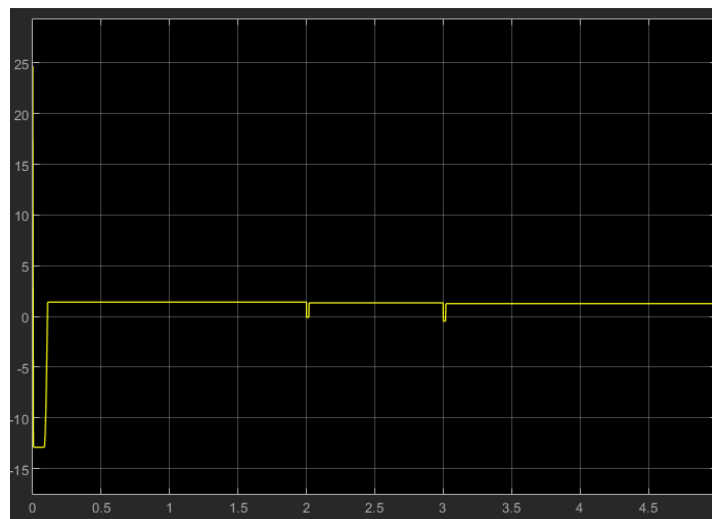


Figure (31) The curvature error plot

Figure (31) shows a consistent steady state error we can deduce that as we drift away the calibration applied voltage 2V the steady state error will increase a bit so from the start we should calibrate the observer at the voltage that provides the desired heating speed.

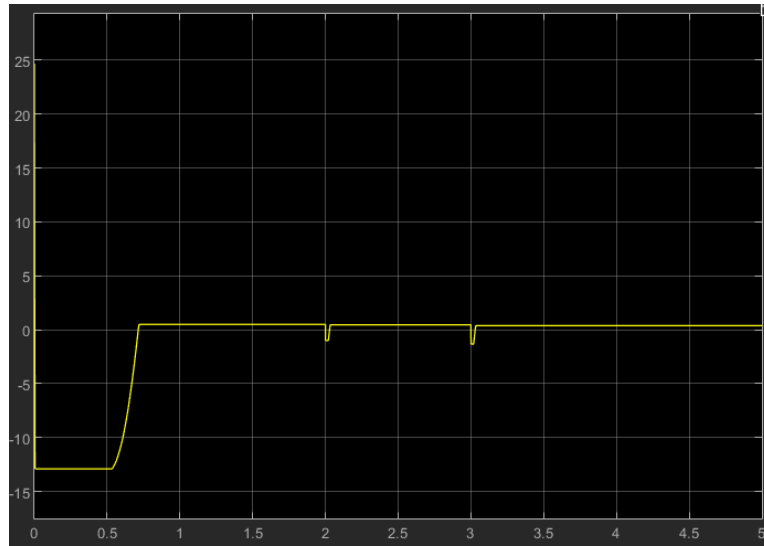


Figure (32) The curvature error plot at 2V applied voltage for the same trajectory

Figure (32) shows the decreased steady state error when I decreased the voltage to the 2V calibration voltage.

5. The conclusion

At first I did discuss the self-sensing procedure and the calibration method to improve our temperature sensing observer, then tested it at the calibration load (zero load) also at applied load case, then proceeded to use the observer to temperature control the system, and do FORC measurement method to extract the feedforward lookup tables (at slightly wrong temperature estimation), then integrate it with the auxiliary outer loop PI controller, and tested the overall system at different load, trajectories and applied voltage to validate how consistent it is against the flaws of the system and the various conditions I spoke about earlier.